Stablecoins & Mainstream Integrations

BUSI 608: Final Presentation

Why can we send a photo across the world instantly, but $100 overseas takes 3 days and costs $10?

Problems with Current Global Monetary Infrastructure

1. High Costs: According to the IMF the average cost of an international transfer is **6.3%**, resulting in **$45 billion per year** are diverted into the hands of intermediaries. That is roughly the same per year cost to end world hunger.

2. Slow Settlement: It can take **1-5 days** to settle a payment and it is subject to business availability/operating times, causing inefficiency in the market and risks to consumers and businesses for **not having access to capital**.

3. Trust Issues: The current domestic and international system has limited visibility into transaction steps, settling statements with **excess fees**, **fraud risk**, longer than stated transaction times, or **incomplete transactions**.

4. Closed Networks: Many are dependent on **private providers** (banks and clearing houses) that offer **limited accessibility** or providers that **require State ID or proofs of residence**. This also includes **political/geopolitical risk** as some counties or persons may be barred from participating in the global or domestic financial system.

5. Limited Ability to Automate: There is often a need for **manual intervention or other settlement frictions** in processing, some systems also have **limited API capabilities & progamabilty capabilities**; this results less flexibility and efficiency.

Stablecoins as a Solution

1. Stablecoins: Digital cash issued as tokens on a blockchain that peg to another market price of an asset.

2. Primary Issuance: Mostly in US Dollars.

3. Market Share: 1% of overall money market circulation

4. Pegging Mechanisms: Tokens are pegged to fiat currencies (e.g., USD) Examples: Tether (USDT), Circle (USDC)

5. Projections: Stablecoins have grown rapidly, with ~$250B issued (including $150B Tether and $60B Circle). Researchers project they could surpass legacy payment rails (SWIFT/BIS handling $5–7T in daily transfers) within the next decade, driven by advantages in speed, cost, and transparency.

PayPal's PYUSD Stablecoin

A fully regulated digital currency backed by traditional assets and issued through established financial infrastructure.

- **NYDFS Regulated**

  Issued by Paxos Trust

- **1:1 USD Redeemable**

  Direct dollar backing

- **Premium Reserves**

  Cash & U.S. Treasuries only

- **No Risk Assets**

  Zero money-market exposure

Reserve Transparency & Verification

Independent Oversight

- Monthly reserve reports published
- KPMG LLP attestations (AICPA standards)
- Available on Paxos PYUSD Transparency Page

July 2025 Attestation Results

1.04B
PYUSD Tokens
Total circulating supply

$1.04B
Reserve Assets
Backing total supply

100%
Fully Coverage
All tokens backed & redeemable

KPMG Attestation Confirmed: All 1,037,124,620 PYUSD tokens are fully backed by $1,040,131,414 in reserve assets, ensuring complete redeemability.

Multi-Layer Protection Framework

Bankruptcy Remote Assets

Customer funds isolated under NY banking law

PayPal Purchase Protection

Eligible purchases covered (limits apply)

Unauthorized Transfer Coverage

Up to **$50,000** lifetime reimbursement cap

PYUSD in PayPal's Ecosystem

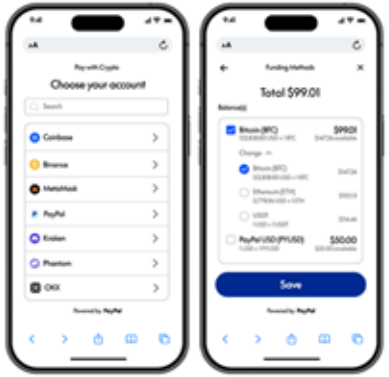
The current features of PYUSD within PayPal's ecosystem are convenient, but still **extremely limited**

- Easy to purchase within PayPal and Venmo apps
- Transfer PYUSD between accounts on these apps with 0 fee, IF they have crypto enabled
- Hold PYUSD (for a 4.0% APY) along with other cryptocurrencies in PayPal or Venmo
- Make international exchanges through XOOM (owned by PayPal)
- Make transfers out of PayPal ecosystem to Ethereum and Solana wallets

PayPal has a bold vision for PYUSD within its ecosystem, but progress remains gradual.

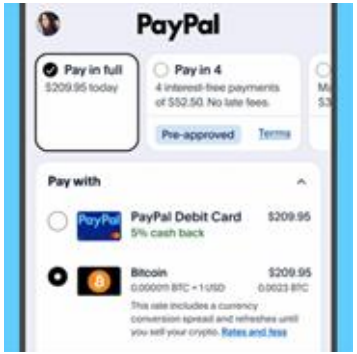
Consumers are able to pay with PYUSD, as PayPal converts it to USD then sends it to the merchant

Announced in July 2025, merchants can now pay other merchants with crypto, including PYUSD



These are features that PayPal has promised, yet they have not been actually implemented.

The question remains whether consumers and merchants have a willingness to adopt to crypto payments



PYUSD's Business Impact

A more efficient and available payment source in the business world

Payment Method	Cost	Settlement Time	Availability	Programmability
PYUSD	**< $0.01**	**Seconds to mins**	**24/7/365**	**Yes**
Wire Transfers	$15-50+	Hours to days	Banking hours	No
Credit/Debit	2-3% + fees	1 - 3 days	24/7	Limited

Interoperability in DeFi Broadening Usage

PYUSD is no longer solely limited to Ethereum, it is now available on Solana, Stellar, and LayerZero-integrated chains like Avalanche, Tron, and Aptos. With an increased presence across multiple blockchains, PYUSD can integrate with a wide range of decentralized applications, making it more versatile and accessible for users without needing to bridge or swap. Allows plug-and-play dynamics with DeFi as it can be used in lending, liquidity pools, yield farming, and automated market makers without customer integration.

LayerZero.

PYUSD's Business Impact

4.0%

Attractive Annual Yield on Holdings for Users

So how does PayPal benefit?

Keeps Users in Ecosystem
Users can send, receive, and hold PYUSD without needing external wallets

Revenue from Reserves
PayPal earns interest income from the assets backing PYUSD

Strengthening Brand Trust
Issuance by Paxos makes PayPal a trusted bridge between TradFi and DeFi

Visa's Stablecoin Strategy

What is their Strategy?

An optional stablecoin payout rail that lets Visa member clients fulfil settlement obligations in stablecoins

Launched March 29, 2021 on Ethereum and expanded on September 5, 2023, to Solana

Visa benefits as it reduces reliance on banking rails, expands global settlement reach and stays competitive vs blockchain-native rails

Operates 24/7/365 which allows merchant payouts in minutes as opposed to days, streamlining the payout process

What is its purpose?

1. **Remittances:** Allows consumers in emerging markets to use digital wallets to send and receive money quickly

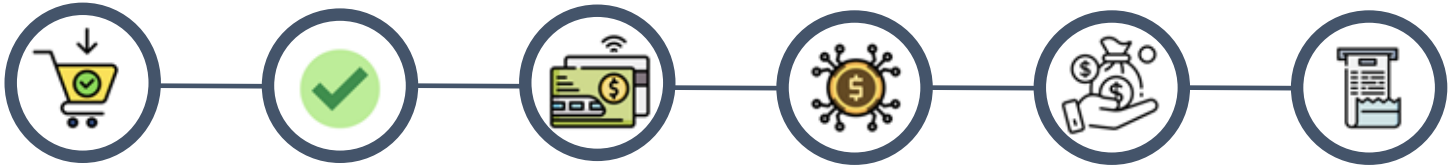
2. **Business Payouts:** Enable fast, cost effective cross-border payouts to creators, contractors and suppliers

3. **Dollar Store of Value:** Give consumers and businesses a dollar-linked place to hold value in more stable assets

4. **Tokenized Assets:** Help banks replace manual processes with smart contracts and build next-gen financial products

Visa’s Stablecoin Strategy

How this process works



Pay at checkout

Card is tapped or entered, Visa/issuer approve the purchase in seconds. No crypto here.

Transactions are cleared

After the sale, Visa batches the merchant’s transactions and calculates what is owed to the acquirer or PSP.

Settlement rail is chosen

If the acquirer has opted in, Visa selects the stablecoin rail (USDC) - otherwise it uses traditional bank rails.

Visa Sends USDC to Acquirer

Visa transfers the exact net amount of USDC to the acquirer’s approved wallet on Ethereum or Solana

Acquirer funds the merchant

The acquirer either forwards USDC to the merchant’s wallet or converts to fiat and sends it to their bank.

Reconciliation and Receipts

Visa and the acquirer match amounts and keep the blockchain transaction records as proof

Comparative Impact: Paypal v. Visa

Similarities:

Features	PayPal	Visa
Speed	Near-instant on blockchain vs. 2-3 day process	Near-instant merchant settlement v. T+2 bank rails
Cost Savings	Lower fees P2P & cross border	Lower FX and wire fees for merchants
Interoperability	PYUSD works on Ethereum & Solana (open networks)	USDC runs on Ethereum & Solana for global payouts
Legitimacy	Regulated issuers & aduits	Partnership with Circle

Differences:

Features	PayPal	Visa
Target User	Everyday consumers & small businesses (retail checkout)	Merchants,acquirers, fintechs (settlement layer)
Business Model	Keep users' funds in PYUSD → earns yields on reserves, and keeps users engaged	Lower settlement costs, keeps card network competitive, collects processing fees
Strategy Scope	Expanding beyond its closed app into open crypto rails	Reinforcing its global payment backbone to stay the default rails for commerce

Why This Matters

- **Complementary, not competing:**
  - PayPal = new user gateway to stablecoins
  - Visa = enterprise highway that moves those stablecoins at scale
- **Network Effect:** as PayPal drives consumer adoption, Visa ensures merchants can actually accept and settle those tokens worldwide

U.S. Regulation of Stablecoins and Risks

1. **Tightening Regulation (Guiding and Setting National Innovation for US Stablecoins- Genius Act)**

- July 2025: law signed to govern cryptocurrency market
- 100% reserve requirement in USD or short-term Treasuries
- Monthly public reserve disclosure mandated
- Marketing rules prevent claims of “government backing”
- Bankruptcy protection extended to stablecoin holders
- Boosts demand for U.S. government securities, lowering borrowing costs and supporting fiscal stability

2. **Trust and Stability:**

- Legislative environment still uncertain
- Peg issues: fixed-value stability can fail without collateral measures
- Run risk if reserves or peg stability are questioned
  Hacker threats: phishing, imposter tokens, DeFi exploits, custodial compromise

3. **Adoption Risks:**

- CBDCs carry geopolitical, financial system, and privacy risks
- Stablecoins in emerging markets (USDC, USDT) may impact local monetary systems
- USDT emphasizes transparency, with regular reserve disclosures

Mitigation Framework

Paxos Regulation + Verifications

Paxos-issued stablecoin PYUSD – along with its reserves – is subject to strict regulatory oversight by the New York State Department of Financial Services.

Diversified Chains + Fraud Tooling

Both Visa and Paypal use a variety of data sets to build a more resilient fraud resistant system. These companies use AI to implement diversified defense.

Clear 1:1 Redemption + Transparency

Reserves are 100% maintained in US dollar deposits, US treasuries, and cash equivalents – that is, customer funds are redeemable 1:1 with Paxos.

Forward Thinking

Tokenization

- Process of representing an asset as a cryptographically secured digital token
- Applicable across industries: real estate, investment funds, and more
- Market size: $24 billion, up 308% in the past 3 years

CBDC Integration

- CBDCs will be critical components of national and cross-border payment infrastructures.
- Continued research and stakeholder engagement with central banks, payment infrastructure operators, financial institutions, technology providers, and researchers are vital to advancing CBDC domestically and internationally

Global Payment Backbone

- It could eventually become the primary method for individuals and businesses to exchange currency quickly and accessible, potentially redefining the financial services industry and significantly impacting investment banks, including their sales and trading desks.

“Stablecoins aren’t just a crypto experiment – they’re becoming the new rails of money.”

THANKS!

Appendix

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